

Sequential EMD-OPD: Representation Alignment before Output Calibration

EMD-OPD

The frozen Bridge enables adaptive cross-layer transport; the resulting checkpoint initializes an unchanged sampled-token OPD stage.

MODULE 0 · ON-POLICY TRAJECTORY

At each stage

Prompt x

Current Student actor
 π_θ

Student-generated response
 $y = (y_1, \dots, y_L) \sim \pi_\theta$

Teacher and Student process the same prompt-response sequence

Losses are evaluated only at valid response-token positions

1 FROZEN BRIDGE

shared low-rank coordinates

FROZEN AFTER FITTING

Projected Student layers

$$Z^S = \{z_i^S\}_{i=1}^{28}$$

Detached Teacher layers

$$\text{sg}(Z^T) = \{\text{sg}(z_j^T)\}_{j=1}^{36}$$

$r \in \{8, 32, 64\}$

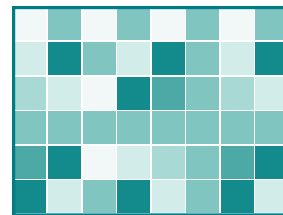
P_S, P_T frozen · all valid response tokens

2 LAYER-LEVEL EMD

representation alignment · Stage 1

Layer cost $C \in \mathbb{R}^{28 \times 36}$

Student i



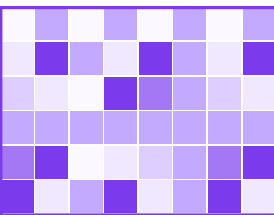
Teacher j

Pairwise response-state cost

$$C_{ij} = \mathbb{E}_b \text{MSE}_{l \in \mathcal{R}_b}(z_{i,t}^S, \text{sg}(z_{j,t}^T))$$

No token×token Gram matrix

Transport plan F^*



mass b_j

Exact balanced OT

$$F^* = \arg \min_{F \geq 0} \langle F, C \rangle$$

$$F \mathbf{1} = a, \quad F^T \mathbf{1} = b$$

$\tau = 1 \cdot \text{exact LP}$

GRADIENT → STUDENT BACKBONE ONLY

DETACH PLAN

EMD representation loss

$$\mathcal{L}_{\text{EMD}} = \lambda_{\text{EMD}} \sum_{i,j} \text{sg}(F_{ij}^*) C_{ij}$$

$\lambda_{\text{EMD}} = 1$

one plan per global mini-batch

Teacher: no_grad

3 HANDOFF

final EMD checkpoint

EMD-aligned Student

step 174

MERGE MODEL

fresh optimizer
+ fresh cosine schedule

4 SAMPLED-TOKEN OPD

output calibration · Stage 2

New on-policy Student trajectory

$$y_t \sim \pi_\theta(\cdot | s_t)$$

Teacher score

$$\log p_T(y_t | s_t)$$

frozen · no_grad

Old Student score

$$\log \pi_{\text{old}}(y_t | s_t)$$

frozen within update

$$A_t = \text{sg}[\log p_T(y_t | s_t) - \log \pi_{\text{old}}(y_t | s_t)]$$

$$\rho_t = \pi_\theta(y_t | s_t) / \pi_{\text{old}}(y_t | s_t)$$

Sampled-token clipped objective → update Student

$$\mathcal{L}_{\text{OPD}} = -E[\min(\rho_t A_t, \text{clip}(\rho_t, 1 - \epsilon, 1 + \epsilon) A_t)]$$

$\epsilon = 0.2$

KEY DESIGN

Stage 1: $\mathcal{L} = \lambda_{\text{EMD}} \mathcal{L}_{\text{EMD}}$ only

CHECKPOINT INITIALIZATION

Stage 2: $\mathcal{L} = \mathcal{L}_{\text{OPD}}$ only

NO JOINT LOSS

Representation alignment supplies initialization; output calibration remains the original OPD code path.